

NB:

- (1) Question No. 1 is compulsory.
- (2) Attempt any three questions out of the remaining five.
- (3) All questions carry equal marks.
- (4) Assume suitable data, if required and state it clearly.

- Q1. ATTEMPT ANY FOUR. [20]**
- a. Design AND gate using Perceptron.
 - b. Suppose we have N input-output pairs. Our goal is to find the parameters that predict the output y from the input x according to some function $y = x^w$. Calculate the sum-of squared error function E between predictions y and inputs x. The parameter w can be determined iteratively using gradient descent. For the calculated error function E, derive the gradient descent update rule $w \leftarrow w - \alpha \frac{dE}{dw}$.
 - c. Explain dropout. How does it solve the problem of overfitting?
 - d. Explain denoising auto encoder model.
 - e. Describe sequence learning problem.
- Q2. a. Explain Gated Recurrent Unit in detail. [10]**
b. What is an activation function? Describe any four activation functions. [10]
- Q3. a. Explain CNN architecture in detail. Suppose, we have input volume of 32*32*3 for a layer in CNN and there are ten 5*5 filters with stride 1 and pad 2; calculate the number of parameters in this layer of CNN. [10]**
b Explain early stopping, batch normalization, and data augmentation. [10]
- Q4 a. Explain RNN architecture in detail. [10]**
b. Explain the working of Generative Adversarial Network. [10]
- Q5 a. Explain Stochastic Gradient Descent and momentum based gradient descent optimization techniques. [10]**
b. Explain LSTM architecture. [10]
- Q6 a. Describe LeNET architecture. [10]**
b. Explain vanishing and exploding gradient in RNNs. [10]

Duration: 3 Hours

[Max Marks: 80]

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- 1 Attempt any **four** **[20]**
 - a) Comment on the Representation Power of MLPs.
 - b) Explain Gradient Descent in Deep Learning.
 - c) Explain the dropout method and its advantages.
 - d) What are Denoising Autoencoders?
 - e) Explain Pooling operation in CNN.
- 2 a) What are the Three Classes of Deep Learning , explain each? **[10]**
 - b) Explain and analyze the architectural of AlexNet Convolution Neural Network. **[10]**
- 3 a) What are the different types of Gradient Descent methods, explain any three of them. **[10]**
 - b) Differentiate between the architecture of LSTM and GRU network. **[10]**
- 4 a) Explain the key components of an RNN. **[10]**
 - b) Consider a CNN layer with the following configuration: **[10]**
 - The input to the layer has 32 channels and a spatial size of 64x64.
 - The convolutional layer has 64 filters (kernels), each of size 3x3, with a stride of 1 and no padding.
 - Each filter is applied to every channel of the input.

Calculate the total number of parameters (weights) in this convolutional layer.
- 5 a) Comment on the significance of Loss functions and explain different types of Loss functions while training a network. **[10]**
 - b) Explain any three types of Autoencoders. **[10]**
- 6 a) What is the significance of Activation Functions in Neural Networks, explain different types Activation functions used in NN. **[10]**
 - b) Explain Generative Adversarial Networks Architecture and its applications. **[10]**

Duration: 3hrs

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- 1 Attempt any **four** [20]
 a) What are Feed Forward Neural Network?
 b) Explain Gradient Descent in Deep Learning.
 c) Explain the dropout method and it's advantages.
 d) What are Undercomplete Autoencoders?
 e) Explain Pooling operation in CNN.
- 2 a) What are the Three Classes of Deep Learning, explain each? [10]
 b) Explain the architecture of CNN with the help of a diagram. [10]
- 3 a) What are the different types of Gradient Descent methods, explain any three of them. [10]
 b) Explain main components of an Autoencoder and it's architecture. [10]
- 4 a) Explain LSTM model, how it overcomes the limitation of RNN. [10]
 b) What are the issues faced by Vanilla GAN models? [10]
- 5 a) What are L1 and L2 regularization methods? [10]
 b) Explain any three types of Autoencoders. [10]
- 6 a) What is the significance of Activation Functions in Neural Networks, explain different types Activation functions used in NN. [10]
 b) What are Generative Adversarial Networks, comment on it's applications. [10]

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- (1) Question No 1 is Compulsory.
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- 1 Attempt any **four** [20]
 - a) Explain basic architecture of feedforward neural network.
 - b) Explain regularization in neural network.
 - c) Explain types of neural network.
 - d) Explain the concept of overfitting and under fitting in neural network.
 - e) Explain basic working of CNN.
- 2 a) Explain the gradient descent algorithm used in neural network. Also discuss types of gradient descent in detail. [10]
b) Explain the working of auto encoders. Also discuss type of auto encoders in detail. [10]
- 3 a) Draw and explain any two modern deep learning architectures. [10]
b) Differentiate between the LSTM and GRU network. [10]
- 4 a) Explain the working of RNN with the help of suitable diagram. [10]
b) Explain how Recurrent Neural Networks (RNNs) are suited for sequential data. Compare the standard RNN architecture with Long Short-Term Memory (LSTM) networks in terms of their ability to handle long-term dependencies. Provide a real-world application where using an LSTM would be significantly more beneficial than a simple RNN and justify your reasoning. [10]
- 5 a) Discuss the role of a loss function in training a neural network. Compare Mean Squared Error (MSE) and Cross-Entropy Loss in terms of their usage, characteristics, and impact on model performance. In which scenarios would using Cross-Entropy Loss be more appropriate than MSE? Justify your answer with a suitable example. [10]
b) Explain architecture of GAN in detail. Also comment on applications of GAN. [10]
- 6 a) What is the significance of Activation Functions in Neural Networks, explain different types Activation functions used in NN. [10]
b) Explain the learning process in a neural network. How does a neural network update its weights during training? Describe the role of forward propagation, loss calculation, backpropagation, and optimization in this learning process. [10]
